

Shrinking an ECDSA Verifier to 898 Bytes

A beginner’s guide to the tricks that worked

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Abstract

This report walks through how an ECDSA/P-256 signature verifier was shrunk from **3076 B** of portable C down to **898 B** of x86-64 machine code — a factor of $3.4\times$. The target context is a boot ROM: the code must fit in a tiny mask-programmed memory, has no libc, and verifies exactly one signature before jumping to firmware. Every byte in a boot ROM is paid for in silicon.

We focus on *why* each trick works, not just *that* it works. No prior exposure to elliptic curves is assumed; we introduce just enough of the mathematics to make sense of the size wins. The full source, test suite, and commit history live in `tv_ecdsa_tiny.S` and the surrounding repository.

Contents

1	The problem	2
1.1	The ground rules	3
1.2	What ECDSA verification actually computes	3
2	The journey at a glance	4
3	Bytecode: call sites are expensive, interpreters are cheap	4
3.1	The problem with calling functions	4
3.2	The fix: a tiny virtual machine	4
3.3	Why 2 bytes, why nibbles	5
4	Dropping Montgomery: the $q = t[\text{top}]$ reduce	5
4.1	What Montgomery multiplication costs you	5
4.2	The special shape of the P-256 moduli	5
4.3	Why that shape makes reduction trivial	6
5	The projective final check: deleting an inversion	6
5.1	Where the mod- p inversion was hiding	7
5.2	Clearing denominators	7
5.3	Folding the OR into a multiply	7
6	bt on n directly: no exponent buffer	7
6.1	The cost of computing $n - 2$	7
6.2	The bit-level observation	8
6.3	The broader lesson	8

7	RCB complete addition: one formula for everything	8
7.1	Why point addition normally has three cases	8
7.2	The complete formula	9
7.3	The demolition	9
7.4	The fine print	9
8	Layout is free, setup is not	10
8.1	Shamir’s trick and its setup cost	10
8.2	Making the source contiguous	10
9	Building constants at runtime	10
9.1	A constant you can compute in the same number of bytes	10
9.2	The knock-on win	11
10	The EFD reschedule: reorder for displacement reach	11
11	Below what looked like the floor	12
11.1	Over-copy: let <code>rep movsq</code> run long	12
11.2	Let the decoder do the handler’s work	12
11.3	Paired big-endian decode	12
12	The micro-grind: x86 encoding folklore	13
12.1	<code>push; pop</code> is a 2-byte <code>mov</code>	13
12.2	Short jumps versus near jumps	13
12.3	Fall-through instead of jump	13
12.4	<code>rep movsq</code> is a free <code>memcpy</code>	13
12.5	Microcode is slow but short	13
12.6	The <code>rcx = 0</code> flow	13
13	Things that didn’t work	14
14	Other limb widths: 11×24, 5×54, 5×56	14
14.1	What a limb is and why its width matters	14
14.2	Quotient estimation, in plain terms	15
14.3	Ninety-six ones in a row	15
14.4	The tracks	15
14.5	Why 8×32 is still ahead	16
15	General principles	16

1 The problem

A boot ROM is a small block of read-only memory burned into a chip at the factory. When the chip powers on, it runs this code first — before any firmware, before any operating system. If the boot ROM verifies the firmware’s signature before handing over control, the chip has a *hardware root of trust*: even an attacker who can overwrite flash cannot run unsigned code.

The catch is size. Mask ROM cells are large compared to flash, every chip on the wafer carries the same bits, and respinning silicon to fix a bug costs months and seven figures. Manufacturers budget boot ROMs in *kilobytes*. Your verifier shares that budget with the flash driver, the clock setup, and whatever else the chip needs to come alive. If you spend 3 KB on signature verification, something else gets cut.

So the game is: **implement ECDSA signature verification over curve P-256, as small as you can make it, without cutting corners on correctness.**

1.1 The ground rules

- **Count everything.** Size is measured on the *object file* — text plus read-only data. If you call `memcpy`, `memcpy` counts. The target here has zero undefined symbols: no `libc`, no compiler runtime, nothing.
- **Correctness is not negotiable.** Every build must pass 607 test vectors: 33 hand-picked edge cases (zero components, out-of-range values, the point at infinity, signature malleability) plus 574 from Google’s Wycheproof suite [5]. A verifier that’s ten bytes smaller but accepts a forged signature is worth exactly nothing.
- **Speed is secondary but not irrelevant.** A boot ROM runs once per power-on, so milliseconds are fine. But there’s a *Pareto frontier*: if build A is both smaller *and* faster than build B, build B is dead. We track both axes.
- **Constant time is not required.** Signature *verification* takes only public inputs — the signature, the public key, the message hash. Nothing secret. Data-dependent branches and variable-time reductions, which would be disqualifying in a *signer*, are fair game here. Several of the tricks below exploit this freedom.

1.2 What ECDSA verification actually computes

You do not need to know why ECDSA is secure to follow this report. You only need to know what the verifier *does*, because that determines where the bytes go.

An ECDSA signature over P-256 is a pair of integers (r, s) , each 256 bits wide. The public key is a point Q on an elliptic curve. Verification, per FIPS 186-5 [1], computes:

$$\begin{aligned}w &= s^{-1} \bmod n && \text{(one modular inversion)} \\u_1 &= e \cdot w \bmod n && (e = \text{the message hash}) \\u_2 &= r \cdot w \bmod n \\(x, y) &= u_1 \cdot G + u_2 \cdot Q && \text{(two scalar multiplications, one point add)} \\&\text{accept iff } x \equiv r \pmod{n}\end{aligned}$$

Here G is a fixed “generator” point baked into the standard, n is the order of G (a 256-bit prime), and p is the field modulus (another 256-bit prime). All the elliptic-curve arithmetic happens modulo p ; the scalars u_1, u_2, w live modulo n .

The heavy lifting is in $u_1 \cdot G + u_2 \cdot Q$. Computing $k \cdot P$ means “add the point P to itself k times” — but since k is a 256-bit number, you do this with a square-and-multiply style loop: 256 iterations of *point doubling* (compute $2P$ from P) interleaved with *point addition* (compute $P + Q$ from P and Q) wherever the scalar has a 1 bit.

Each point doubling or point addition is in turn a fixed sequence of field operations: multiplications, squarings, additions, subtractions, all modulo p . A P-256 point addition is roughly a dozen field multiplications plus change. One standard trick: instead of working with (x, y) pairs directly, the loop carries an extra coordinate Z and represents the running point as (X, Y, Z) with $x = X/Z$. Divisions are expensive (see §5); this *projective* representation defers them all to one division at the very end.

So the byte budget is roughly: one field multiplier (mod p and mod n), one point-add and one point-double formula that call it, one scalar-multiply loop that calls *them*, one inversion, and the top-level glue that decodes the inputs and checks the result. Constants for p, n, G , and the curve coefficient b take another 128–160 bytes.

2 The journey at a glance

Table 1 gives the headline numbers. The rest of this document goes through each row in enough detail that you could, in principle, rediscover the trick yourself.

Size	Δ	What changed
3076 B		Portable C, gcc -Os, 32-bit limbs
2873 B		First hand-written x86-64 assembly
1712 B	-1161	§3: Bytecode interpreter
1427 B	-285	fast.S : Montgomery + mulx + Shamir’s trick
1397 B	-30	Micro-grind (register allocation, fall-through)
1300 B	-97	§4: Drop Montgomery — the $q = t[\text{top}]$ reduce
1195 B	-105	§5: Projective final check — no mod- p inversion
1105 B	-90	§6: bt on n directly — no exponent buffer
1012 B	-93	§7: RCB complete addition — one formula, no branches
985 B	-27	§8: Addend slot shift — layout is free, setup is not
942 B	-43	§9: Build p at runtime + fe_iszero inlines
933 B	-9	§10: EFD reschedule — reorder for disp8 reach
898 B	-35	§11: Over-copy, decoder preload, paired decode

Table 1: Major milestones. Sizes between rows are connected by dozens of smaller wins; see §12 for the general flavour. The rule marks where tiny.S forked from fast.S: rows above it optimised for speed first; rows below trade speed for size.

3 Bytecode: call sites are expensive, interpreters are cheap

From **2873 B** to **1712 B**.

3.1 The problem with calling functions

The first hand-written assembly version looked the way you’d expect: a function for field-multiply, a function for field-add, a function for field-subtract, and then pt_dbl and pt_add calling them in sequence.

On x86-64, calling a three-argument function costs about 15 bytes:

```
lea    rdi, [r14 + 32*DST]    ; 4 bytes (or 7 if disp > 127)
lea    rsi, [r14 + 32*SRC1]   ; 4 bytes
lea    rdx, [r14 + 32*SRC2]   ; 4 bytes
call   fe_mul                 ; 5 bytes
```

Point doubling is about 8 field operations; point addition is about 12. The verifier’s final-check sequence is another 10 or so. That’s roughly 60 call sites $\times$ 15 bytes each: **900 bytes spent on nothing but “load three pointers and jump.”**

3.2 The fix: a tiny virtual machine

Notice that every one of those call sites has the same *shape*: one destination, two sources, one operation. And all the operands live in a contiguous block of 32-byte field elements on the stack. So instead of emitting 15 bytes of setup per call, we encode each call as **2 bytes of data** and write a small loop that decodes and dispatches:

```

; Each instruction: byte0 = (s2<<4)/op, byte1 = (dst<<4)/s1
; With 16 slots and 16 ops, everything fits in nibbles.
bc_run:
    lodsw                ; fetch 2 bytes of bytecode into ax
    ; ... decode nibbles into rdi/rsi/rdx (about 30 bytes) ...
    movzx ecx, byte [r12 + rax] ; jump-table lookup (op index -> handler offset)
    add rcx, r12
    call rcx              ; dispatch
    jmp bc_run

```

The dispatch loop is about 60 bytes. Each handler (multiply, add, subtract, ...) is another 10–40 bytes — but you already had those as functions. The bytecode streams themselves are $60 \times 2 = 120$ bytes.

Net effect: ~900 bytes of call sites becomes ~180 bytes of interpreter plus bytecode. The `pt_dbl` function literally *disappears* as native code — it is now 25 bytes of data in `.rodata`.

3.3 Why 2 bytes, why nibbles

You could encode each op in 3 bytes (one per operand) and have room for 256 slots. But you don't *have* 256 slots — point addition uses perhaps a dozen temporaries. Four bits per operand, packed two to a byte, is the right granularity. And four bits of opcode gives you 16 operations — the final build uses ten.

There's a subtle win hiding here: **the interpreter doesn't care which *specific* 16 slots you use**. You can reassign slots to make other code shorter. We will exploit this ruthlessly later (§8).

4 Dropping Montgomery: the $q = t[\text{top}]$ reduce

From **1397 B** to **1300 B**. Cornerstone of the size-optimised build.

4.1 What Montgomery multiplication costs you

The fast implementation used *Montgomery form* [4]: numbers are stored as $a \cdot R \bmod m$ where $R = 2^{256}$. This makes the reduce-after-multiply step very fast — but it carries hidden fixed costs:

- A precomputed constant $m_0^{-1} = -m^{-1} \bmod 2^{64}$ for each modulus (p and n): **16 B** of data plus the code to pass it into the multiplier on every call.
- A precomputed constant $R^2 \bmod p$ to convert inputs *into* Montgomery form: **32 B** of data plus a TOMONT bytecode op.
- Conversion *out* of Montgomery form at the end: more bytecode ops.

None of these are big individually, but they add up to about 60–80 bytes — and for size, we'd rather have a slower multiplier with no baggage.

4.2 The special shape of the P-256 moduli

Here is where we get lucky. Write out p and n in hex and stare at the top word:

```

p = ffffffff 00000001 00000000 00000000 00000000 ffffffff ffffffff ffffffff
n = ffffffff 00000000 ffffffff ffffffff bce6faad a7179e84 f3b9cac2 fc632551

```

Both moduli have a top 32-bit word of 0xFFFFFFFF. And both satisfy $2^{256} - m < 2^{224}$ — that is, both are “just barely less than” 2^{256} .

4.3 Why that shape makes reduction trivial

After a schoolbook multiply, you have a 512-bit product t and you want $t \bmod m$. The textbook approach is trial division: estimate the quotient, multiply, subtract, correct. Quotient estimation is where the complexity lives.

But if the top word of m is 0xFFFFFFFF, quotient estimation becomes **free**. Split t into 32-bit words $t[0..15]$ and slide a 9-word window from the top down. At each position j :

1. Let $q = t[j+8]$ — just *read* the top word of the window. No division.
2. Subtract $q \cdot m$ from the window.

Why is $q = t[j+8]$ a good quotient estimate? Because m ’s top word is $2^{32} - 1$, the true quotient $\lfloor t_{\text{window}}/m \rfloor$ is within 1 of $\lfloor t_{\text{window}}/2^{256} \rfloor = t[j+8]$.

Concretely: write the window as $W = q \cdot 2^{256} + L$ with $0 \leq L < 2^{256}$. Then

$$W - q \cdot m = q(2^{256} - m) + L < 2^{32} \cdot 2^{224} + 2^{256} = 2 \cdot 2^{256},$$

so after one subtraction the window is at most two moduli wide, and its new top word — $\lfloor (W - qm)/2^{256} \rfloor$ — is either 0 or 1. If it’s 1, one more pass brings it to 0. Two passes per position, worst case.

```
; Reduce: slide a 9-dword window from j=8 down to j=0.
; At each j: q = t[j+8]; t[j..j+8] -= q*m.
; The SAME 26-byte primitive handles both schoolbook multiply
; AND this reduce --- they're both "mul ebx; accumulate into [rdi]".
.Lreduce:
    mov     ebx, [rdi + 32]    ; q = top dword of current window
    ; ... call the shared mul-accumulate primitive ...
    sub     rdi, 4             ; slide window down one dword
    dec     ecx
    jnz     .Lreduce
```

The payoff: no m_0^{-1} , no R^2 , no conversion ops. Field elements are stored in plain form the whole time. The reduce code is *shorter* than Montgomery’s, and one 26-byte inner primitive serves double duty for multiply and reduce.

The trade-off is speed: this runs at 32-bit granularity (8×8 word multiplies instead of 4×4), so it’s about $2 \times$ slower. For a boot ROM, we’ll take that trade every time.

A subtlety worth noting. The $q = t[\text{top}]$ trick requires the top 32-bit word to be all-ones. The top 64-bit word of p is 0xFFFFFFFF00000001, which is *not* all-ones. So even after we later upgraded the schoolbook multiply to 64-bit limbs (**+14 B** for about a $2 \times$ speedup), the reduce stayed at 32-bit granularity. This is why `tv_ecdsa_tiny.S` has a slightly odd split: 64-bit multiply, 32-bit reduce.

5 The projective final check: deleting an inversion

*From **1216 B** to **1195 B**. One of those rare wins that’s both smaller and faster.*

5.1 Where the mod- p inversion was hiding

The ECDSA check (last line of §1.2) is “accept if $x \equiv r \pmod{n}$,” where x is the affine x -coordinate of the computed point. But the scalar multiplication loop doesn’t produce affine coordinates — it produces *projective* coordinates (X, Y, Z) , where the affine x is recovered as $x = X/Z$ (for homogeneous projective) or $x = X/Z^2$ (for Jacobian).

“Divide by Z ” in a finite field means “multiply by Z^{-1} .” For prime p , Fermat’s little theorem gives $Z^{p-1} \equiv 1 \pmod{p}$, so $Z^{-1} \equiv Z^{p-2}$: inversion is exponentiation. That exponentiation is a 256-iteration square-and-multiply loop — another ~ 380 field multiplications on top of everything else.

The straightforward verifier does this inversion, recovers x , and compares it to r . That inversion call, its argument setup, and the final compare-and-reduce-mod- n block cost about 45 bytes.

5.2 Clearing denominators

Here’s the algebra trick. Instead of computing $x = X/Z$ and checking $x \stackrel{?}{=} r$, **clear the denominator**: check $X \stackrel{?}{=} r \cdot Z \pmod{p}$. No inversion needed.

There’s one wrinkle. The ECDSA check is modulo n , but X and Z live modulo p , and $p \neq n$. Since $0 \leq x < p < 2n$, there are at most two integers congruent to $r \pmod{n}$ in that range: r itself, and $r + n$ (the latter only when $r + n < p$). So the full check becomes:

$$X \equiv r \cdot Z \pmod{p} \quad \text{OR} \quad X \equiv (r + n) \cdot Z \pmod{p}.$$

When $r \geq p - n$ — a roughly 2^{-128} sliver of the r range — the sum $r + n$ wraps around mod p and the second disjunct tests a spurious value. That’s harmless: forcing the scalar multiplication to land on a *specific* spurious point is as hard as the discrete log problem the whole scheme rests on.

5.3 Folding the or into a multiply

Checking an OR in assembly usually means a branch, and branches are byte-expensive. But we’re working modulo a prime p , and in a field, **a product is zero iff one of its factors is zero**. So let

$$d_1 = X - r \cdot Z, \quad d_2 = X - (r + n) \cdot Z,$$

and check whether $d_1 \cdot d_2 \equiv 0 \pmod{p}$. One multiply, one zero-check, no branches.

The whole thing — compute $r \cdot Z$, compute d_1 , compute d_2 , multiply, check-zero — is 7 field operations, which is 14 bytes of bytecode. The 45 bytes of native inversion-and-compare are gone, replaced by 14 bytes of data.

And as a bonus, we no longer call the inverter with modulus p at all. The inverter now only ever runs mod n (for s^{-1}). That removes an INV_P handler from the bytecode jump table and lets us hardcode n into the inverter, saving another handful of bytes (§6).

6 bt on n directly: no exponent buffer

From [1124 B](#) to [1105 B](#).

6.1 The cost of computing $n - 2$

Fermat’s little theorem gives $s^{-1} \equiv s^{n-2} \pmod{n}$. The exponentiation is a 256-step square-and-multiply loop that reads one bit of the exponent $n - 2$ per iteration: square always, multiply if the bit is 1.

The exponent is $n - 2$. Previously we stored n as a constant, and at runtime: copy n to a 32-byte stack buffer, subtract 2 from the buffer, then have the loop read bits from the buffer with the `bt` instruction.

That’s an `enter` to allocate the buffer, a `rep movsq` to copy, a `sub` to decrement, and a `leave` to clean up: 33 bytes of machinery just to turn n into $n - 2$.

6.2 The bit-level observation

Write out the low byte of n and $n - 2$ in binary:

$$\begin{aligned} n \bmod 256 &= 0x51 = 0101\ 0001 \\ (n - 2) \bmod 256 &= 0x4F = 0100\ 1111 \\ \text{XOR} &= 0x1E = 0001\ 1110 \end{aligned}$$

Subtracting 2 borrows from bit 1. That bit is 0 in n , so the borrow propagates: through bit 2 (also 0), through bit 3 (also 0), until it reaches bit 4 — which is 1 and absorbs it. **Bits 5 through 255 of $n - 2$ are identical to bits 5 through 255 of n .**

So: don’t compute $n - 2$ at all. Read bits directly from the n constant that’s already sitting in `.rodata`. For bits 5 and up, n gives the right answer. For bits 0–4, patch in a 7-byte special case:

```
cmp    ebx, 4
jb     .Lmul      ; bits 0-3: n-2 has 1111 here, force multiply
je     .Lnext     ; bit 4:   n-2 has 0, n has 1, force skip
bt     [r12 + 0N], ebx ; bits 5+: n and n-2 agree, read n directly
jnc    .Lnext
.Lmul:
; ... multiply step ...
```

The 33-byte buffer machinery becomes 7 bytes of branching. Six bytes of the original machinery survive (the `init` copy still has to run), and the `bt` operand grows by one byte (it now reaches into `.rodata` instead of `[rsp]`). Net: $-33 + 7 + 6 + 1 = -19\text{B}$.

6.3 The broader lesson

This trick is *ludicrously specific* to the bit pattern of the P-256 group order. It would not work for most other primes. But that’s exactly the point: when optimising for size, there is no “generic” — you stare at the *actual* constants and ask what accidents of their structure you can exploit.

7 RCB complete addition: one formula for everything

From [1071 B](#) to [1012 B](#). The biggest single-commit win after the bytecode interpreter.

7.1 Why point addition normally has three cases

The classical formulas for adding two elliptic curve points $P + Q$ have a dirty secret: they divide by $x_P - x_Q$. When $P = Q$ (doubling), that’s 0/0. When $P = -Q$ (the result is the point at infinity $\mathcal{O}$), same problem. And if either input is already $\mathcal{O}$, the formulas don’t even apply — $\mathcal{O}$ has no coordinates.

So a traditional `pt_add_acc` routine starts with a three-way branch:

```
if (acc.z == 0)    { acc = Q; return; }    // acc was infinity
// ... run first half of add formula to compute H = x_Q - x_P ...
```



```

if (H == 0) {
    if (Rr == 0)    { pt_dbl(acc); return; } // P == Q, use doubling
    else           { acc = infinity; return; } // P == -Q
}
// ... finish the addition formula ...

```

In our bytecode world this was 62 bytes of native branching, plus *three* separate bytecode streams: `bc_dbl` (50 B), `bc_add1` (24 B, the pre-check half), and `bc_add2` (28 B, the post-check half). Total: 164 bytes.

7.2 The complete formula

In 2015, Renes, Costello, and Batina [2] published a point addition formula that **has no exceptional cases**. One sequence of 43 field operations gives the correct answer for *every* pair of inputs: distinct points, equal points, inverse points, and the point at infinity. No branches. No special cases. It just works.

The formula uses *homogeneous* projective coordinates: the affine point is recovered as $(X/Z, Y/Z)$. Two triples name the same point whenever they differ by a common nonzero scale factor: $(X : Y : Z)$ and $(\lambda X : \lambda Y : \lambda Z)$ are the same thing, which is what the colons signal. Infinity is the class with $Z = 0$, which on this curve forces $X = 0$ too: any $(0 : Y : 0)$ with $Y \neq 0$ will do. Feeding the formula $(0 : 1 : 0) + Q$ really does produce Q , and $(0 : 1 : 0) + (0 : 1 : 0)$ really does produce $(0 : 1 : 0)$ — the edge cases are handled by the algebra, not by code.

7.3 The demolition

With RCB, `pt_add_acc` becomes:

```

pt_add_acc:                ; 11 bytes. Was 62.
    ; Addend already in slots 5,6,7. Just dispatch bc_rcb.
    xor    edi, edi        ; bytecode offset 0 = bc_rcb
    jmp    bcrun_off

```

Doubling is not a separate operation any more. To double the accumulator, copy it into the addend slots and call the *same* routine — $P + P = 2P$, and RCB handles that case like any other.

The three bytecode streams (**102 B** total) collapse into one 43-operation stream (**88 B**). The 62-byte branch tree becomes an 11-byte dispatch. Setup bookkeeping for the new Z coordinate eats **+6 B** of that back; net **−59 B**, and a whole category of bugs (forget to handle one edge case, accept a forged signature) structurally eliminated.

7.4 The fine print

RCB does more arithmetic than the classical “generic add” — 43 operations versus about 16. So each addition is about $2.5\times$ slower. For a boot ROM, we shrug. For a speed-optimised build, you’d keep the branchy version.

One invariant turned out to be load-bearing: the RCB formula never *writes* to the slots holding Q . It reads them, but the outputs go elsewhere. This means the scalar-multiply loop doesn’t need to save and restore Q across each add — Q just sits there, undisturbed, for all 256 iterations. That invariant saved another dozen bytes of copy-in-copy-out in the loop.

8 Layout is free, setup is not

From **1001 B** to **985 B**.

The bytecode interpreter addresses 16 slots by a 4-bit index. Which slot is “slot 5” is completely arbitrary — it’s a nibble in a data table. Renumbering slots costs **zero bytes** in the bytecode. But slot numbers show up in *native* code too, as memory displacements — and those are not free.

8.1 Shamir’s trick and its setup cost

Shamir’s trick [6, §14.88] (due originally to Straus [7]) computes $u_1G + u_2Q$ with a single 256-iteration loop instead of two: at each bit position, double once, then conditionally add G , conditionally add Q . It halves the number of doublings.

The trick needs backup copies of both G and Q that survive the whole loop, because the “working” addend slots get overwritten. After the RCB switch, each point is 3 slots (X, Y, Z). So the setup is: copy G (3 slots) and Q (3 slots) to backup slots 16–21.

The naive setup was three separate copies: G ’s X, Y from one place, a borrowed $Z = 1$ from another, Q from a third. About 28 bytes of `lea/rep movsq` triples.

8.2 Making the source contiguous

What if the six source slots were *already contiguous* before the copy? Then the setup is one `rep movsq` with count 24. Ten bytes instead of 28.

To make this happen: (a) arrange for the early validation bytecode to leave slots 2–7 holding exactly $G_x, G_y, 1, Q_x, Q_y, 1$ in that order; (b) shift the RCB addend from slots 4,5,6 to slots 5,6,7 so it lines up.

Part (a) costs 2 extra bytes of bytecode (one more “copy 1 here” op). Part (b) is a **pure nibble permutation**: swap some 4-bit fields in `bc_rcb`, zero-byte cost. Net: **−16 B**.

The general principle: data layout changes inside the bytecode are free; exploit that freedom to make the native code shorter.

9 Building constants at runtime

From **948 B** to **942 B** — but the direct saving is zero. The entire win is a knock-on effect.

9.1 A constant you can compute in the same number of bytes

The field prime p for P-256 is

$$p = 2^{256} - 2^{224} + 2^{192} + 2^{96} - 1.$$

As 32 bytes of `.rodata` it’s **32 B**. But look at the hex:

```
ffffffff 00000001 00000000 00000000 00000000 ffffffff ffffffff ffffffff
```

Seven of the eight 32-bit words are either `0x00000000` or `0xFFFFFFFF` — only `0x00000001` is neither. Writing this out one dword at a time with `stosd` (which stores `eax` and auto-advances `rdi`) takes **17 B**:

```
or    eax, -1      ; eax = FFFFFFFF    (3 B)
stosd                ; dword 0          (1 B each; stosd = AB)
stosd                ; dword 1
stosd                ; dword 2
```

```

xor    eax, eax        ; eax = 0          (2 B)
stosd                     ; dword 3
stosd                     ; dword 4
stosd                     ; dword 5
inc     eax             ; eax = 1          (2 B)
stosd                     ; dword 6
neg     eax             ; eax = FFFFFFFF   (2 B)
stosd                     ; dword 7

```

That looks like a **-15 B** win — but the constant didn’t sit in a vacuum. Deleting it means the places that *used* it need new plumbing: a pointer register to hold the built-at-runtime address, a new bytecode op so validation can check coordinates against it, save/restore of the new register in the prologue. About **+15 B** of overhead, netting out to roughly zero.

So why bother? Because **where the 32 bytes disappeared from matters more than how many were saved.**

9.2 The knock-on win

The bytecode jump table stores handler addresses as *8-bit* offsets (one byte per handler). This means every handler must be within 255 bytes of the table base. The largest handler (`fe_inv_m`) was sitting at offset 247 — only 8 bytes of headroom before the whole scheme breaks and every table entry doubles in size.

The `cP` constant lived between the jump table and the handlers. Deleting it from `.rodata` moved every handler 32 bytes closer to the table. Suddenly there’s 36 bytes of headroom — enough to inline a small `fe_iszero` helper into its only caller, for **-6 B**. That is the entire saving in the section header.

(We had already done the related trick for the curve coefficient b , computed *in bytecode* from the curve equation itself: since G is on the curve, $b = G_y^2 - G_x^3 + 3G_x \pmod{p}$. Four field ops, 8 bytes of bytecode, replacing **32 B** of constant — a direct saving this time, because bytecode costs no plumbing.)

Size optimisations interact: freeing space in one region can unblock a seemingly unrelated win elsewhere. Keep a list of “things I’d do if only I had 10 more bytes of headroom at X .”

10 The EFD reschedule: reorder for displacement reach

From **935 B** to **933 B**.

x86-64 memory operands with a small displacement ($-128 \leq d \leq 127$) encode in one byte; larger displacements take four. One `lea rdi, [r14+160]` in the hot copy path was paying the three-byte tax because slot 5 is $5 \times 32 = 160$ bytes from the slot base, and $160 > 127$.

If we had a register that already pointed near slot 5, we could use a short displacement from that register. And `r8` was already holding `&cP` — but `cP` was at slot 22, way too far.

So: move cP. If `cP` lived at slot 8 instead ($8 \times 32 = 256$ from base), then slot 5 would be at `r8 - 96`, which fits in a signed byte. But slot 8 was being used as scratch by the RCB formula.

The Explicit-Formulas Database [3] lists the RCB operations in a particular order, but the order is not sacred — only the *data dependencies* are. By hoisting two independent steps earlier in the sequence, Z_1 dies sooner, slot 2 becomes reusable as a sixth temporary, and slot 8 is *never written* by `bc_rcb`. It can hold `cP` for the entire scalar multiplication.

Shuffling bytecode nibbles is free. The displacement shrinks from **+160** (4 bytes) to **-96** (1 byte). Minus one byte for a setup adjustment elsewhere, net **-2 B**.

Two bytes is small, but it illustrates how far the rabbit hole goes: *instruction scheduling in a data table to change register allocation to shrink an addressing mode in unrelated native code.*

11 Below what looked like the floor

From **933 B** to **898 B**.

The 933-byte build sat unchanged for long enough that it started to feel like a hard limit for this architecture. It wasn't. None of the following changed the algorithm — still the same $q = t[\text{top}]$ reduce, same RCB formula, same bytecode — but 35 more bytes came out of the encoding and the layout.

11.1 Over-copy: let `rep movsq` run long

Several places copy a 3-slot point (12 quadwords) with `rep movsq` and then immediately need `rdi` pointing somewhere further along for the next operation. The old pattern: copy 12 quadwords, then `lea rdi, [...]` to reposition — 4 to 7 bytes of address arithmetic.

Since `rep movsq` advances both `rsi` and `rdi` by the count, copying *more* than needed does the repositioning for free. If the slots between the real destination and the next needed position are scratch — written before they're next read — they can soak up the extra writes without consequence. Doubling a 12-qword copy to 24 qwords costs nothing (`mov c1,24` is the same 2 bytes as `mov c1,12`) and deletes the `lea`.

This is the dataflow equivalent of writing past the end of your buffer on purpose, because you own the memory on the other side and know nobody's looking.

11.2 Let the decoder do the handler's work

The bytecode dispatch loop (§3) already does arithmetic on every fetch: it unpacks three nibbles and turns each into a slot pointer via `base + nibble×32`. That multiply-and-add runs regardless of which opcode follows.

Some handlers were doing their own address arithmetic on entry — computing a secondary pointer from a primary one, or offsetting into a structure. But a slot nibble is four free bits, and a slot pointer is a free multiply-by-32. If you can encode the handler's desired offset as a slot number, the decoder has already computed it by the time the handler runs. The handler shrinks; the bytecode byte it reads from was already there.

The pattern is general: whenever a fixed-cost decode stage sits in front of variable handlers, ask what the decode stage could compute on the handlers' behalf.

11.3 Paired big-endian decode

The verifier has to parse four 32-byte big-endian integers from the input blob: r and s from the signature, Q_x and Q_y from the public key. They arrive in pairs — (r, s) adjacent in one buffer, (Q_x, Q_y) adjacent in another — and land in adjacent slots.

Each decode was a 5-byte `call fe_from_be`, so each pair cost 10 bytes of call sites. A 5-byte helper that calls `fe_from_be` once and then falls through into it a second time turns each pair into one 5-byte call. Two pairs: **+5 B** for the helper, **-10 B** at the call sites, net **-5 B**.

12 The micro-grind: x86 encoding folklore

Between the big structural wins, hundreds of bytes came out one or two at a time. These don't make for a good story individually, but the patterns repeat.

12.1 `push; pop` is a 2-byte `mov`

`mov rax, rbx` is 3 bytes. `push rbx; pop rax` is 2 bytes (for registers that don't need a REX prefix). It clobbers 8 bytes of stack below `rsp`, which is fine if you're not using the red zone for anything else.

This pattern appears about a dozen times in the final code. It saved roughly **-10 B** total.

12.2 Short jumps versus near jumps

A jump to a target within ± 127 bytes encodes as 2 bytes (`EB xx`). Beyond that, it's 5 bytes (`E9 xx xx xx xx`). Every function-layout decision — what order the handlers go in, whether a helper sits before or after its caller — affects which jumps are short.

Several commits in the history are just “swap these two blocks so this `jmp` goes from 5 bytes to 2.” Three bytes per swap, maybe five such swaps over the project: **-15 B**.

12.3 Fall-through instead of jump

Even better than a short jump is no jump. If function A always ends by calling function B, and nothing else sits between them, delete the `call` from A and let execution fall through. `Fadd` falls through into the conditional-subtract epilogue; `Nmul` overwrites the modulus pointer and falls through into `Fmul`. Each fall-through is **-2 B** to **-5 B**; four or five over the project comes to roughly **-15 B**.

12.4 `rep movsq` is a free `memcpy`

`rep movsq` is 3 bytes and copies `rcx` quadwords from `[rsi]` to `[rdi]`. If you can get `rsi`, `rdi`, and `rcx` into the right state as a side effect of surrounding code — `lodsb` leaves `rsi` advanced, `stosq` leaves `rdi` advanced, loop counters naturally end at zero — then a block copy is nearly free.

The final code has about eight `rep movsq` instructions and not a single explicit copy loop. The art is in arranging the dataflow so the registers are already right.

12.5 Microcode is slow but short

The `loop` instruction (decrement `rcx` and jump if nonzero) is 2 bytes. The equivalent `dec rcx; jnz` is 5 bytes. But `loop` is *microcoded* on modern x86 and costs several extra cycles per iteration — Agner Fog's tables [8] put it at 6 on Skylake, worse on AMD Zen.

If the loop runs ten times, nobody cares. If it's the inner loop of the field multiplier and runs 950,000 times per signature, that's millions of wasted cycles. The `-DSMALL_MUL8` build takes the size win; the default build spends **+14 B** to run at full speed. Both are on the Pareto frontier.

12.6 The `rcx = 0` flow

`mov cl, 8` is 2 bytes and sets the low 8 bits of `rcx`. If the upper 56 bits happen to already be zero — because a previous `rep` instruction counted down to zero, or a previous `loop`

exited — you just saved 3 bytes over `mov ecx, 8` (5 bytes) or 1 byte over `push 8; pop rcx` (3 bytes).

This is a *global* invariant: “`rcx` is zero at these twelve points in the control flow, and the code downstream relies on it.” It’s fragile — reordering two blocks can silently break it — and it saved perhaps **−8 B** across the whole binary. The source has comments at every point where the invariant is consumed.

13 Things that didn’t work

Not every idea pans out. A few instructive failures:

Merging schoolbook and reduce into one loop.

The multiply loop and the reduce loop have similar shapes — walk an array, multiply by something, accumulate. Merging them would share the loop machinery. It also overflows: the accumulated carry during the multiply phase can exceed what the reduce phase expects, and you get wrong answers on about one input in 2^{30} . Reverted after a Wycheproof failure.

Zero-skip in the scalar loop.

Half the scalar bits are zero, so skipping the add on zero bits should be a speedup. It was 40% *slower*: the branch mispredicts roughly half the time (the bits are effectively random) and the penalty swamps the saved work. `perf stat` showing branch-miss rate was the diagnostic that killed this idea.

Solinas reduction for both moduli.

The prime p has a beautiful sparse form ($2^{256} - 2^{224} + 2^{192} + 2^{96} - 1$) that admits a reduction with no multiplies at all, just shifts and adds. But n doesn’t have that form, so you need *two* reducers instead of one. For a speed build that’s worth it (`-DSOLINAS_P` exists); for the size floor, one generic reducer serving both moduli is smaller.

14 Other limb widths: 11×24, 5×54, 5×56

Everything above is one architecture: eight 32-bit words per field element, plain (non-Montgomery) reduction. That choice was made early and then optimised hard. But it’s a choice, and other choices carve up the problem differently. This section sketches the alternatives explored in parallel — none has beaten **898 B** yet, but each taught something about where the bytes actually go.

14.1 What a limb is and why its width matters

A 256-bit integer doesn’t fit in a machine register, so it’s stored as an array of K smaller pieces — *limbs* — of W bits each, with $K \cdot W \geq 256$. The slack $K \cdot W - 256$ is headroom for carries to accumulate between reductions. The main build uses $K = 8$, $W = 32$: no slack, carries propagated immediately.

Multiplying two K -limb numbers by the schoolbook method is K^2 single-limb multiplies — for $K = 8$ that’s 64; for $K = 5$ it’s 25. Fewer limbs means dramatically less arithmetic. But fewer limbs means wider limbs, and wider limbs have wider products: a 54×54 bit multiply overflows a 64-bit register and needs 128-bit intermediate results, which on x86-64 means the two-register `rdx:rax` output of `mul` and extra bookkeeping to handle it. A 24×24 bit multiply fits comfortably in 64 bits with room to accumulate a column’s worth of partial products before anything overflows.

So the trade-off runs: small K saves multiplications but pays for wide-product handling; large K keeps each step simple but does many more of them. Where the minimum lands

depends on the instruction set, the reduction strategy, and — it turns out — a bit-level accident in the constant p .

14.2 Quotient estimation, in plain terms

Before getting to the accident, it helps to see what all reduction strategies are really doing. After multiplying two numbers modulo p , you have a result up to p^2 — roughly twice as many bits as you want — and need to bring it back below p . Conceptually this is division: compute $t \bmod p$ by finding how many times p goes into t and subtracting that many copies.

Division is what you want to avoid. It’s the slowest arithmetic instruction on almost every chip ever made, and it doesn’t pipeline. Every practical reduction scheme replaces the division with a *guess*: estimate the quotient cheaply, subtract, see how close you got, correct if needed. The schemes differ in how they guess.

The $q = t[\text{top}]$ reduce (§4) guesses by reading off the top word — which works because p ’s top word is all ones, so p is within a hair of the next power of two and “how many times does p go in” is almost exactly “how many times does 2^{256} go in,” which is just a shift. It’s the same reason long division by hand is easy when the divisor is 999: you barely have to think about the trial digit.

Montgomery reduction [4] guesses from the other end. It asks: what multiple of p can I add to t so that the *bottom* W bits become zero? Then those zero bits are shifted off, and the value shrinks by W bits per round. Finding that multiple normally takes one extra multiply per round — you compute $q = (t \bmod 2^W) \cdot c$, where the constant $c = -p^{-1} \bmod 2^W$ is precomputed once and baked into the binary. One multiply by a fixed constant per limb, which is why the main build dropped Montgomery: that multiply and its constant are overhead.

14.3 Ninety-six ones in a row

Here is the accident. Recall $p = 2^{256} - 2^{224} + 2^{192} + 2^{96} - 1$. That trailing -1 borrows all the way up to bit 96 before the $+2^{96}$ term stops it — so bits 0 through 95 of p are all ones. You can read it straight off the hex from §4: the bottom three words are `ffffffff ffffffff ffffffff`.

What does this do to the Montgomery constant $c = -p^{-1} \bmod 2^W$? If $W \leq 96$, then $p \bmod 2^W$ is a W -bit block of ones — that is, $p \equiv -1 \pmod{2^W}$. Then $p^{-1} \equiv -1$ as well (since -1 is its own inverse), and $c = -(-1) = 1$.

The per-limb Montgomery multiply is by 1. It disappears. For any limb width up to 96 bits, Montgomery reduction mod p costs no more per round than the $q = t[\text{top}]$ scheme does — the guess is free in both. Every limb width on the table except 8×32 in its non-Montgomery form exploits this: the Montgomery machinery that §4 threw out comes back, but without the part that made it expensive.

(The order n doesn’t share this structure — its low bits are `fc632551` — so the mod- n inverter still pays for its Montgomery constant. But inversion runs once; the field multiplier runs thousands of times.)

14.4 The tracks

Track	K	W	Reduction strategy
8×32	8	32	non-Montgomery, $q = t[\text{top}]$
11×24	11	24	Montgomery, $c_p = 1$, 64-bit accumulators
5×54	5	54	Montgomery, $c_p = 1$, 128-bit products
5×56	5	56	Montgomery, $c_p = 1$, byte-aligned decode

11×24 . Eleven limbs of 24 bits gives 264 bits of capacity — 8 bits of slack per field element. A 24×24 product is 48 bits; even after summing eleven of them down a column, the total stays under 64 bits. No `rdx:rax` splitting, no 128-bit adds, just plain `imul` and `add`. The multiplier inner loop is the simplest of any track. The cost: $11^2 = 121$ multiplies per field operation versus $8^2 = 64$, almost double, and the carry-propagation code that runs after each operation is longer because there are more limbs to walk.

5×54 . Twenty-five multiplies. The K^2 scaling wins decisively on arithmetic count. But 54-bit limbs mean 108-bit products, handled as `rdx:rax` pairs, and the reduction involves 128-bit-wide shift-and-add chains. And 54 doesn't divide 8: decoding a big-endian input into 54-bit limbs requires loading a word, shifting by a variable amount, and masking — a bit-twiddling loop that the byte-granular tracks don't need.

5×56 . Same $K = 5$ as above, so the same 25 multiplies and the same $c_p = 1$. The difference is that 56 bits is exactly 7 bytes. Decoding a 56-bit limb from a byte buffer is an unaligned 8-byte load followed by an 8-bit mask — no variable shift loop, no bit-offset bookkeeping. Two extra bits of width per limb buys back a chunk of the parsing machinery for nearly nothing, and $56 < 96$ so the Montgomery constant is still 1.

This track is the newest and the least-ground-down. Whether the byte-aligned decode saves more than the slightly wider products cost is not yet settled.

14.5 Why 8×32 is still ahead

The 8×32 track has one structural advantage none of the others share: a single reducer serves both moduli. Both p and n have a top 32-bit word of `0xFFFFFFFF`, so the $q = t[\text{top}]$ scheme handles both with one code path. Every Montgomery track needs the $c_p = 1$ trick for the field and something else for the order — two reducers, or one reducer with a mode switch.

That, plus a substantial head start in accumulated encoding tricks, keeps 8×32 in front. But the lead is not structural inevitability; it's accumulated effort. The other tracks have tricks the main build can't use — Montgomery cancellations that only exist when there are R -factors to cancel — and some of those haven't been fully mined.

15 General principles

If you're trying to shrink your own binary, here's what transfers:

1. **Count call sites, not functions.** A function called once is as expensive as its call site plus its body. A function called sixty times is sixty call sites plus one body. The interpreter trick (§3) wins because it collapses sixty 15-byte call sites into sixty 2-byte data entries.
2. **Algebra beats branching.** The projective check (§5) and the RCB formula (§7) both replace control flow with arithmetic. Arithmetic is data; branches are code. Data is cheaper.
3. **Your constants are not random.** The `bt-on- $n$` trick (§6), the $q = t[\text{top}]$ reduce (§4), and building p at runtime (§9) all exploit specific bit patterns in the P-256 constants. Stare at your constants in hex. Stare at them in binary. There is structure there.
4. **Layout is a degree of freedom.** Slot numbering (§8), function order (short jumps), constant order (block copies), rodata-before-text (`push imm8`) — none of these affect

semantics, all of them affect size. Change them last, after the architecture settles, because they're brittle.

5. **Track both axes.** It's easy to micro-optimize yourself into a corner where you've saved 5 bytes and lost $10\times$ speed. Plot size against cycles for every checkpoint; if your new point is dominated by an old one on both axes, you've wasted your time.
6. **Tests before bench.** Every single one of the ~ 130 commits in this project passed 607/607 test vectors before being committed. Several ideas were killed by tests (see above). A smaller verifier that accepts forgeries is negative progress.

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